

Effect of Parent Rocks on Chemical and Mineralogical Properties of Some Oxisols in Brazil¹

N. CURI AND D. P. FRANZMEIER²

ABSTRACT

A toposequence of soils developed primarily from basalt in the Central Plateau of Brazil and a pedon developed primarily from itabirite (a hematite-quartz metamorphic rock) were studied in the Quadrilátero Ferrífero. The itabirite-derived Oxisol consists almost entirely—900 to 950 g kg⁻¹ (90–95%)—of Fe oxides and oxyhydroxides, compared with 250 to 300 g kg⁻¹ in the basalt-derived Oxisol. The soil on itabirite also has coarser particle size, higher particle density, and somewhat lower surface area. It lacks kaolinite and gibbsite but contains more goethite and has greater magnetic susceptibility. It also is much lower in all essential plant nutrients (except Fe and P), and much lower in several non-nutrient minor elements as well. Magnetite grains in basalt are relatively abundant in minor elements which could be released slowly during mineral weathering. The effect of the parent material is still evident even in highly weathered soils, indicating the need for the separation between itabirite- and basalt-derived Oxisols in a system of soil classification. In itabirite-derived soils the Fe₂O₃/TiO₂ ratio (H₂SO₄ digestion) is >16 and the magnetic susceptibility of the whole soil is >189 × 10⁻⁶ m³ kg⁻¹ (15 000 × 10⁻⁶ cgs). These limits of soil properties reflect the original parent material and could be the differentiating characteristics in soil classification systems. Particle density, amount of trace elements, and capacity to fix phosphorus are accessory properties associated with the differentiating characteristics.

Additional Index Words: goethite, hematite, basalt, itabirite, minor elements.

Curi, N., and D.P. Franzmeier. 1987. Effect of parent rocks on chemical and mineralogical properties of some Oxisols in Brazil. *Soil Sci. Soc. Am. J.* 51:153–158.

THE DUSKY RED LATOSOLS OF BRAZIL by definition have 2.5YR or redder hues and contain more than 18% Fe₂O₃ by sulfuric acid extraction. They developed from mafic rocks such as basalt and most of these soils are highly productive. Soils developed from itabirite are very similar to these Dusky Red Latosols in many properties but are much less productive. Itabirite is a hematite-quartz-schist, a metamorphic rock very high in Fe content which is mined as Fe ore in various places. This difference in productivity could be due to the chemical and mineralogical properties of the soils. The objectives of this study were to determine the influence of parent rocks, basalt and itabirite, on the chemical and mineralogical properties of Oxisols developed from them, and to suggest diagnostic criteria to separate them in soil classification systems.

MATERIALS AND METHODS

Soils

The basalt-derived soils were sampled in a toposequence in southern Goiás State, near the Minas Gerais border (Curi and Franzmeier, 1984). Mainly, the soil in the summit position of the toposequence, G6, will be discussed because its topographic position is most like the itabirite-derived soil. The soil from basalt supports cerrado, a savanna-like veg-

etation, and has an ustic soil moisture regime and an isohyperthermic soil temperature regime. Samples of the Bo horizon were collected at a depth of 1 m. The basalt rock sample was collected from a nearby well. The soil formed from itabirite, M1, was sampled in the western part of the Quadrilátero Ferrífero, near the Serra da Moeda, a few meters from the road between Belo Horizonte and Rio de Janeiro, in Minas Gerais State. Regionally the area is a pediment at the foot of a structural ridge (Serra da Moeda), but in the local landscape the sampling site is on an upland section between small drainageways diverging to an inner valley. The natural vegetation is cerrado. The soil has an udic soil moisture regime and a thermic soil temperature regime. Samples of the Bo horizon were collected at a depth of 0.5 m, and the itabirite rock sample was collected deeper in the same pit.

The possibility that material other than the rocks sampled for reference influenced the soils must be recognized. For example, the basalt rock contained no quartz, but soil G6 did contain it (Curi and Franzmeier, 1984). This quartz must have originated from some kind of transported sediment. However, the soil over itabirite contained 90 to 95% Fe₂O₃, so the influence of transported material on this soil must have been minimal. From field evidence and the data presented here we believe that the underlying material had a major, but not exclusive, influence on soil properties so, for brevity in the discussion, we will refer to basalt-derived soils and itabirite-derived soils.

Soil G6 is classified as dystrophic, moderate A, very clayey, Dusky Red Latosol, basaltic substrate in the Brazilian classification and clayey, oxidic, isohyperthermic, Acrustox in *Soil Taxonomy* (Soil Survey Staff, 1975). Soil M1 is a dystrophic, prominent A, medium texture, Ferriferous Latosol, itabiritic rock substrate and a fine-loamy, ferritic, thermic, Haplic Acrorthox.

Laboratory Methods

Total Chemical Analysis by ICP—The contents of major elements, as well as some trace elements, of finely-ground rock and soil were determined by inductively coupled plasma-atomic emission spectrometry (ICP) with a Jarrell-Ash Model 975 Plasma AtomComp direct reader by the Indiana Geological Survey Laboratory (Lechler et al., 1980). A reagent blank was used for the low standard and granite (G2G2) and diabase (DIAB) reference samples from the U.S. Geological Survey were used for the high standards for calibrating the instrument. Two or three replicate determinations were made of each sample. Dynamic background and interelement corrections were made via inboard computer, and the corrected results were expressed as weight percent.

Total Chemical Analysis by Atomic Absorption—Cobalt and Mo were analyzed by atomic absorption using the samples prepared for ICP analysis and a Varian AA6D instrument.

Total Chemical Analysis by PIXE—Proton bombardment was utilized to induce characteristic x-rays in all elements present in samples prepared as thin films of finely-ground rock or soil material. The method involves observation and counting of these x-rays thereby providing a non-destructive

¹ Contribution from the Agron. Dep., Purdue Univ. Agric. Exp. Stn., West Lafayette, IN 47907. Journal Paper no. 10,632. Received 21 Feb. 1986.

² Former Graduate Fellow on leave from Departamento de Ciência do Solo, Escola Superior de Agricultura de Lavras, Minas Gerais State, Brazil (MEC/CAPES program); and Professor, respectively.

tive, quantitative measurement of the concentration of each element (with $Z > 12$, except Co and Mo) in the samples. The procedure was conducted by the Physics Dep., Purdue Univ. (Kullerud et al., 1979).

Phosphorus Sorption—A 2-g whole soil sample was shaken for 12 h with a series of 40-mL P solutions (0–200 ppm) prepared in 0.01 M CaCl_2 . In order to inhibit microbial growth, two drops of toluene were previously added. The samples were centrifuged for 30 min. at 2000 rpm and the P in the supernatant was determined by the method of Murphy and Riley (1962). The amount of P adsorbed was calculated from the difference between the initial concentration of P in solution minus the final concentration of P in solution. The adsorption data were evaluated assuming two types of adsorption sites and applying the Langmuir two-surface equation (Syers et al., 1973; Rajan and Fox, 1975). The samples were then shaken for 12 h with 40 mL of 0.01 M CaCl_2 , centrifuged for 30 min. at 2000 rpm, and the P in the supernatant was determined. The amount of P desorbed was taken as that present in the solution after shaking.

Differential X-ray Diffraction Analysis For Fe Oxides (DXRD)—A modification of the method for obtaining differential XRD patterns of soil Fe oxides (Schulze, 1981) was used by incorporating 1 μm α -alumina (corundum) as an internal standard (Bryant et al., 1983). The fine clay fraction was selected for analysis. The amount of Al substitution in goethite was calculated by the method of Schulze (1984) and in hematite by the method of Steinwehr (1967). Mean crystallite dimension was calculated by the Scherrer equation (Klug and Alexander, 1974).

Other laboratory procedures were described by Curi and Franzmeier (1984).

RESULTS AND DISCUSSION

Physical Characteristics

The results of the particle size distribution, particle density, specific surface, and moisture retention analyses are in Table 1. The clay fraction predominates in the soil from basalt (G6) but the sand fraction predominates in the soil from itabirite (M1). The interdependence between the parent material composition and the soil texture is evident because coarse-sized quartz, which is extremely resistant to weathering, is present in itabirite and absent in basalt. Also the silt/clay ratio is higher for M1 soil than for G6. The soil from basalt has much higher values for clay content, specific surface, moisture retained at -1.5 MPa, and moisture equivalent (Table 1), than the one from itabirite (M1).

The soil formed from itabirite rocks (M1) has a higher particle density (3.57) than the one developed from basalt (2.74) in accordance with their heavy/light mineral ratios. This could be considered as a criterion to separate basalt- and itabirite-derived soils in the Brazilian soil classification.

Chemical and Mineralogical Characteristics

Routine Chemical Analyses

The most notable chemical characteristic is that the soil from itabirite consists almost entirely of Fe oxides. Several CBD extractions removed 588 g Fe kg^{-1} , which is equivalent to 84% Fe_2O_3 or 93% FeOOH . Later in this paper it will be shown that the total Fe content is probably somewhat greater and that the Fe oxide minerals contain some Al substituted for Fe.

Table 1. Physical and chemical properties of <2 mm samples of the B₀ horizons.

Soil property	Units	G6 soil (Basalt)	M1 soil (Itabirite)
Depth	m	1.0	0.5
Total sand	g kg^{-1}	170	600
Silt	g kg^{-1}	120	130
Clay	g kg^{-1}	710	270
Particle density	Mg m^{-3}	2.74	3.57
Specific surface area	$\text{m}^2 \text{g}^{-1}$	150	131
Moisture at -1.5 MPa	g kg^{-1}	241	93
Moisture equivalent	g kg^{-1}	263	118
pH (1:2.5) in H_2O		5.4	5.8
pH (1:2.5) in 1 N KCl		5.2	5.6
Exch. Ca + Mg + K + Na	cmol (+) kg^{-1}	0.25	0.14
Exchangeable Al	cmol (+) kg^{-1}	0	0
Base saturation	%	9	4
Organic C	g kg^{-1}	7.9	9.4
Available P	mg kg^{-1}	1	1
H_2SO_4 dissolution			
Si	g kg^{-1}	84	4
Al	g kg^{-1}	151	29
Fe	g kg^{-1}	184	442
Ti	g kg^{-1}	21.3	1.9
P	g kg^{-1}	0.7	0.9
Mn	g kg^{-1}	0.5	0.3
$\text{SiO}_2/\text{Al}_2\text{O}_3$	(molar)	1.07	0.28
$\text{SiO}_2/(\text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3)$	(molar)	0.67	0.03
$\text{Al}_2\text{O}_3/\text{Fe}_2\text{O}_3$	(molar)	1.71	0.13
Al_d (CBD dissolution)	g kg^{-1}	15	29
Fe_d (CBD dissolution)	g kg^{-1}	167	588
Phosphate adsorption, max.	mg P kg^{-1}	1345	1015
Phosphate desorption, avg.	mg P kg^{-1}	21.9	35.7

Both of these factors would result in a calculated content of Fe oxide minerals of more than 95% of the soil.

The pH determined in water and 1 M KCl, exchangeable cations, organic C, available P, and the total oxide data from sulfuric acid digestion can be found in Table 1.

Although both soils have very low base saturation and are dystrophic ($< 50\%$ base saturation at pH 7.0), the soil from basalt has a higher base saturation than the soil from itabirite, reflecting the different parent rocks. The available P is extremely low in both soils, probably due to the "fixation" of this element mostly by Fe oxyhydroxides.

The soil derived from basalt has 184 g Fe kg^{-1} and 21.3 g Ti kg^{-1} from H_2SO_4 digestion data ($\text{Fe/Ti} = 9$) while the M1 soil (From itabirite) has 442 g Fe kg^{-1} and 1.9 g Ti kg^{-1} ($\text{Fe/Ti} = 232$). In other soils from Brazil (EMBRAPA, 1985), neither the Fe nor the Ti contents alone differentiate between soils formed from basalt and those formed from itabirite. However, a $\text{Fe}_2\text{O}_3/\text{Ti O}_2$ ratio of 15.2 seems to constitute the upper limit for Dusky Red Latosols (Latosol Roxo-LR). Thus, in the Brazilian soil classification system now being developed, a $\text{Fe}_2\text{O}_3/\text{Ti O}_2$ ratio > 16 ($\text{Fe/Ti} > 19$) could be used to distinguish between Oxisols formed from mafic rocks and those from itabirite rocks.

The $\text{SiO}_2/\text{Al}_2\text{O}_3$ and $\text{SiO}_2/\text{R}_2\text{O}_3$ molar ratios (Ki and Kr values) are quite low because of loss of Si from basalt and itabirite, and originally low content of Si and high content of Fe in itabirite.

Extraction with CBD removes mainly free oxides of Fe and Al, but H_2SO_4 extraction dissolves most of the clay fraction and some secondary mineral con-

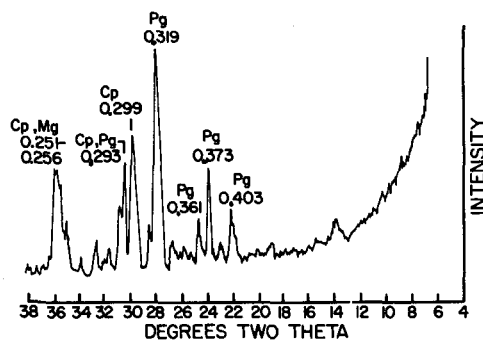


Fig. 1. X-ray diffractogram of the fresh basaltic rock (random powder). Pg = plagioclase, Cp = clinopyroxene (augite), Mg = magnetite; numbers represent d-spacings in nm.

stituents of aggregates or particles coarser than 2 μm . A comparison of the results of these extractions shows that most of the Fe and Al in soil M1 and most of the Fe in soil G6 are present as free oxides. However, much of the Al in soil G6 was not extracted, probably because it is in gibbsite and kaolinite. The H_2SO_4 extraction appears to underestimate the Fe extracted from the M1 soil because it is much lower than the Fe removed by CBD.

Phosphorus Adsorption-desorption Studies

The results of this analysis are presented in Tables 1 and 2. The Oxisol from basalt (G6) has higher P adsorption values, and higher surface area, than the one from itabirite (M1), which has more free Fe. The amount of Fe_d does not appear to be a determining factor for P adsorption, which is also mentioned by Bigham et al. (1978).

The data included in Table 2 indicate the existence of at least two adsorption sites with very distinct adsorption characteristics. Region 1 corresponds to a population of sites with relatively low adsorption maximum (b_1) and high binding energy constant (k_1), while region 2 corresponds to another population of sites with larger adsorption maximum (1.14–1.56-fold) but lower binding energy constant (5.08- to 6.90-fold). This observed trend is in agreement with the data found by Syers et al. (1973) and Rajan and Fox (1975).

The results for P desorption (Table 1) show low values for both soils, but the M1 soil had higher values than the G6 soil. The main conclusion of the sorption study is that the basalt-derived Oxisol B horizon (71% clay) adsorbed more and desorbed less P than the itabirite-derived Oxisol B horizon (27% clay).

Rock Mineralogy

The x-ray diffractogram of the fresh basaltic rock (Fig. 1) showed a dominance of plagioclase (labradorite) and clinopyroxenes (augite); magnetite was also present. Quartz was not detected.

Itabirite (hard and soft rock) (Fig. 2) contained goethite, magnetite (or maghemite), hematite, and quartz. Concretions from a pedon similar to M1 had hematite, maghemite, and quartz, but not goethite (Fontes et al., 1985a). Comparing the hard and soft rock x-ray diffractograms, goethite peaks are more intense (0.417 nm diffraction line) whereas magnetite (0.295 nm) and hematite (0.367 nm) peaks are less intense in the soft

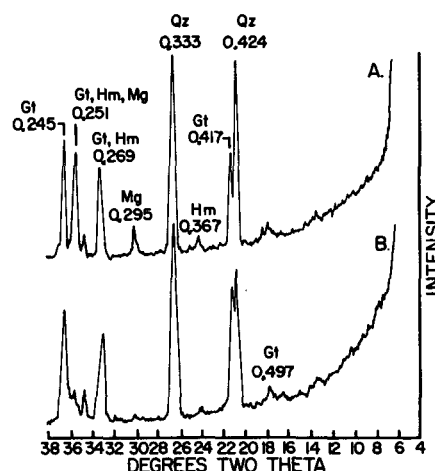


Fig. 2. X-ray diffractograms of the hard (A) and soft (B) itabirite rock (random powder). Gt = goethite, Hm = hematite, Mg = magnetite (or maghemite), Qz = quartz; numbers represent d-spacings in nm.

rock. The color of the soft itabirite rock is much browner than the hard rock which is consistent with its higher goethite content. Conversion (by hydration) of the Fe oxides to goethite in itabirite rocks has been reported (Gair, 1962; Dorr and Barbosa, 1963; Simmons, 1968). However, Schwertmann and Taylor (1977) reported that hematite cannot be converted to goethite by simple hydration. Goethite must be dissolved and hematite reprecipitated for this conversion to occur.

Coarse Fraction Mineralogy

In the light minerals, quartz is the predominant mineral, whereas in the case of heavy minerals, opaque minerals were dominant in both soils. Both the magnetite grain counts and the magnetic susceptibility data of the sand and whole soil fractions (Table 3), are greater in the M1 soil, suggesting that some interdependence is involved.

Clay Mineralogy

The mineralogy of the clay fraction in the basalt-derived Oxisol is dominated by kaolinite, gibbsite, hematite, goethite, and maghemite (Table 3). Anatase is also present.

The mineralogy of the clay fraction of the Oxisol from itabirite rock (M1) is dominated by goethite, hematite, and maghemite. The x-ray diffractograms

Table 2. Phosphate adsorption parameters (b_1 and k_1 values) of the soils studied, as estimated from graphical resolution of "Eadie-Hofstee" plots.

Adsorption parameter	Units	G6 soil (Basalt)	M1 soil (Itabirite)
b_1	mg P/kg soil	525	475
b_2	mg P/kg soil	820	540
b_t	mg P/kg soil	1345	1015
$b_2:b_1$		1.56	1.14
k_1	ml/ μg P	16.25	6.90
k_2	ml/ μg P	3.20	1.00
k_t	ml/ μg P	19.45	7.90
$k_1:k_2$		5.08	6.90

$\dagger b$ = adsorption maximum.

$\ddagger k$ = binding energy constant.

Table 3. Soil color, mineralogy of the fine clay fraction (<2 μ m), magnetite content of the heavy portion of the very fine sand fraction, and magnetic susceptibility of the B₀ horizons.

Characteristic	Units†	G6 soil (Basalt)	M1 soil (Itabirite)
Moist soil color		2.5YR 3/4	10R 3/4
Kaolinite (DTA)	g kg ⁻¹	220	<10
Gibbsite (DTA)	g kg ⁻¹	220	<10
Peak area from DXRD			
Goethite (110)	CPS $\times$ WHH	5.0	18.1
Hematite (012)	CPS $\times$ WHH	8.2	5.7
Maghemite (206:220)	CPS $\times$ WHH	3.1	3.8
Al substitution			
Goethite	mol mol ⁻¹	0.24	0.23
Hematite	mol mol ⁻¹	0.16	0.16
Mean crystallite dimension			
Goethite (111)	nm	24.6	27.0
Hematite (110)	nm	38.0	25.3
Magnetite grain count	% (hv)	60	80
Magnetic susceptibility			
Sand	$\times 10^{-6}$ m ³ kg ⁻¹	48	267
Fine clay	$\times 10^{-6}$ m ³ kg ⁻¹	28	133
Whole soil	$\times 10^{-6}$ m ³ kg ⁻¹	40	283

† Abbreviations: CPS, counts per second; WHH, width of peak at half-height; % (hv), number of magnetite grains per 100 very fine sand heavy mineral grains.

did not show any evidence for kaolinite, gibbsite, or anatase. This mineralogical composition is also reflected in the high Fe content and low Si, Al, and Ti contents of the M1 soil. Most Al probably was consumed by replacement of Fe in the goethite and hematite structures. However, IR absorption (not shown) indicated the presence of small amounts of gibbsite.

Differential x-ray diffraction analysis for Fe oxides (DXRD) showed that the M1 soil contains relatively more goethite and less hematite and has smaller hematite crystals compared with the G6 soil (Table 3). Goethite and hematite had Al-substitution of about 0.24 and 0.16 mol mol⁻¹, respectively, in both soils. In the M1 soil this value for the fine clay is probably much higher than the Al substitution in coarser fractions of the Fe oxides because the amount of Al extracted from the whole soil with CBD or with H₂SO₄ was 29 g kg⁻¹ (Table 1), which is only about 5% of the CBD-extractable Fe + Al.

The soil formed from itabirite contains more goethite in relation to hematite and is redder than the soil formed from basalt. Usually soils that contain more hematite are redder (Davey et al., 1975; Resende, 1976; Bigham et al., 1978; Childs et al., 1979; Schwertmann et al., 1982). The itabirite soil is different from most of those studied previously in that much of the goethite in the solum was inherited from the parent material. Possibly the pigmenting effect of inherited goethite, much of which is in the sand and silt fractions (Table 1), is less than that formed in place. Maghemite probably formed by the aerial oxidation of magnetite (Schwertmann and Taylor, 1977; Coey, 1985).

For the Fe-rich M1 sample, the intensities of hematite, goethite, and maghemite XRD peaks before and after CBD treatment were measured and compared. These data show that CBD dissolved all the original hematite and maghemite but only about 88% of the goethite. These results agree with the observation of Mehra and Jackson (1960) that goethite dis-

Table 4. Elemental composition of rock and soil samples (<2 mm) and common ranges for soils.

Element	Meth- od†	Basalt	G1 soil	G6 soil	Itabirite	M1 soil	Common range in soils‡
Major elements							
g kg⁻¹							
Al	I	68	139	142	2	26	10-300
	P	90	139	193	<36	<36	
Ca§	I	62	0.42	0.33	0.16	0.16	7-500
	P	58	0.48	0.45	0.56	0.43	
Fe§	I	101	126	153	497	616	7-550
	P	92	114	174	427	577	
K§	I	8.9	0.38	0.34	<0.08	<0.08	0.4-30
	P	7.6	0.18	0.18	<0.15	<0.15	
Mg§	I	30.4	0.67	0.62	0.78	0.24	0.6-6
Mn§	I	5.68	0.49	0.62	1.03	0.39	0.02-3
	P	1.22	0.44	0.60	1.16	0.39	
Na	I	17.3	0.22	0.14	0.09	0.13	0.75-7.5
P§	I	2.54	1.92	2.16	1.50	2.53	0.2-5
	P	2.02	<0.8	<0.8	<0.8	0.80	
Si	P	236	124	130	146	54	230-350
Ti	I	18.9	42.6	43.9	0.02	2.95	1-10
	P	17.0	39.5	46.9	0.11	3.30	
Minor elements							
mg kg⁻¹							
Ag	P	1.8	8.6	7.1	4.2	<1.8	0.01-5
As	P	<2	6.4	9.6	2.3	35.7	1-50
Ba	P	370	<66	73	66	84	100-3000
Co	A	65	78	105	64	99	1-40
Cr	P	69	220	300	<52	<52	1-1000
Cu§	P	141	157	149	<6	18	2-100
Ga	P	18	41.1	49.5	2.7	20.8	5-70
Ge	P	<1.2	<1.2	2.1	<1.2	4.6	1-50
Mo§	A	<0.01	<0.01	0.21	<0.01	<0.01	0.2-5
Nb	P	14	46.6	49.2	<11	11	
Ni	P	90	110	95	<39	<39	5-500
Pb	P	7.9	9.4	13	<9	<9	2-200
Rb	P	21	<3.4	<3.4	<3.4	<3.4	50-500
Sb	P	10	7.5	9.2	<7.5	35	
Se	P	<1.5	<1.5	<1.5	<1.5	3.2	0.1-2
Sr	P	478	8.6	16.3	14	19	50-1000
U	P	<2.9	7.7	14	40.1	90.9	
V	P	400	869	970	<86	<86	20-500
Y	P	29	18.7	20.5	11	5.3	25-250
Zn§	P	59.3	<1.0	3.2	<1.0	<1.0	10-300
Zr	P	187	562	566	<132	132	60-2000

† A = atomic absorption, I = ICP, P = PIXE.

‡ Lindsay, 1979.

§ Essential for plant growth.

solved more slowly than hematite in CBD treatment. Also, Norrish and Taylor (1961) reported that high-Al content goethite is less soluble than low-Al content goethite in dithionite extract.

The magnetic susceptibility data for the sand and whole soil (Table 3) reflect mostly the higher magnetite content in the sand fraction of the M1 than the G6 soil as determined by grain counts. Apparently all size fractions of the M1 soil have higher magnetic susceptibility. The susceptibility in the clay fraction is probably due to maghemite. The much greater magnetic susceptibility of the soil from itabirite suggests that this property, in addition to the Fe/Ti ratio, could be used to separate the two kinds of soils in a soil classification system. A limit of 189×10^{-6} m³ kg⁻¹ ($15\,000 \times 10^{-6}$ cgs) is suggested for consideration. Data for itabirite soils could be compared with those for other Latosols from Brazil (Resende et al., 1986). To further test this suggestion, magnetic susceptibility can be readily estimated in the field (Santana, 1984) so this criterion could be adapted for use by field soil scientists.

Total Chemical Analyses

The results of the chemical analyses are presented in Table 4 and are compared with the common range of concentration in soils. The results from the two analytical methods for the major elements showed only general agreement. Both methods were calibrated within concentration ranges of elements normally found in the samples analyzed by that method, but some of the elemental concentrations of the soil samples may have been outside of this range. One of the standards used in the ICP procedure was a diabase sample. Because diabase is chemically similar to basalt, the basalt rock analyzed and the soil on basalt should be similar to that standard for most major elements. The G6 soil should be compared with M1 because they are in summit topographic positions. Data for the Bo horizon (1-m depth) of the G1 soil, in a footslope position, are included to provide information about variability within a landscape.

For minor elements the results by PIXE and atomic absorption methods are reported. Some of the elements were also analyzed by ICP (Curi, 1983). Some of these analyses agreed with PIXE results and some were greatly different. If a particular element was much higher in our samples than in the rock samples used to calibrate the ICP instrument the extrapolation might be too great. ICP analysis gave very high results for Cd. Because of the importance of this element in human health, further analysis of soils for this element should be made.

Relative to soil formation, it is interesting to note which elements are more plentiful in the soil than in the rock from which it formed and which are less plentiful (Table 5). In the basalt-derived soil, the alkali and alkaline earth cations, Mn, and Si are less abundant in the soil, which most likely represents a net loss of these elements. This trend was less apparent in the soil from itabirite because this rock has very low contents of most of these elements. Elements more abundant in both soils than in the rock were Al, Ti, Fe, and some transition metals. This probably represents a loss of other constituents rather than gains of these elements.

The M1 soil from itabirite is very high in Fe content, and according to ICP analysis, it contains 616 g Fe kg⁻¹—which is the equivalent of 88% Fe₂O₃ or 98% FeOOH.

Considering the elements for which analyses are reported in Table 4, all essential plant nutrient elements except Fe and P are more abundant in soil G6 than in soil M1. For both soils, Ca, K, Mg, and Mo are near or below the lower limit of the common range in soils (Lingsay, 1979); Cu is above the upper limit in soil G6 and Fe is above it in soil M1. These results are similar to a broader survey of soils and plants in southern Brazil (Sillanpää, 1982) which concluded: "Only the contents of Fe are at the 'normal' international level. The content of B is somewhat low, that of Mo is very low while those of Cu and Mn are very high." Also, the generally high level of Cu and low level of Mo might cause problems in plant nutrition because Cu interferes with the role of Mo in the enzymatic reduction of NO₃ (Kabata-Pendias and Pendias, 1984).

Table 5. Changes in elemental concentration in soils in relation to the rocks from which they formed.

	G6 soil from basalt	M1 soil from itabirite
Increase†	Al, Ti, Fe; Ag, As, Co, Cr, Ga, Ge, Mo, Nb, Pb, Th, U, V, Zr	Al, Ti, Fe; As, Co, Cu, Ga, Ge, Sb, Se, Th, U
Decrease‡	Ca, K, Mg, Mn, Na, Si; Ba, Rb, Sr, Y, Zn	Mg, Mn, Si; Ag

† Increase—concentration in soil is $>1.5 \times$ (concentration in rock); decrease—concentration in soil is $<0.67 \times$ (concentration in rock).

‡ Fe showed a large increase, but did not meet the $1.5 \times$ requirement because of its abundance in the rock.

The elemental composition of the soils largely reflects the concentration of these elements in the parent rocks. The minor elements may have been largely associated with magnetite grains. Magnetite grains were analyzed by energy dispersive x-ray analysis (EDAX) which showed that those from basalt had higher concentrations of many trace elements than those from itabirite. This trend is corroborated by the very low values obtained for trace elements in concretions from a pedon similar to M1 (Fontes et al., 1985b). Thus, even in these highly weathered soils the higher content of minor elements in basalt compared with itabirite is reflected in the soils developed from these rocks.

REFERENCES

- Bigham, J.M., D.C. Golden, S.W. Buol, S.B. Weed, and L.H. Bowen. 1978. Iron oxide mineralogy of well-drained Ultisols and Oxisols: II. Influence on color, surface area, and phosphate retention. *Soil Sci. Soc. Am. J.* 42:825–830.
- Bryant, R.B., N. Curi, C.B. Roth, and D.P. Franzmeier. 1983. Use of an internal standard with differential X-ray diffraction analysis for iron oxides. *Soil Sci. Soc. Am. J.* 47:168–173.
- Childs, C.W., B.A. Goodman, and G.J. Churchman. 1979. Application of Mossbauer spectroscopy to the study of iron oxides in some red and yellow/brown soil samples from New Zealand. p. 555–565. *In* M.M. Mortland and V.C. Farmer (ed.) *Int. Clay Conf.*, 6th, Oxford, England, 1978. Elsevier, Amsterdam.
- Coe, J.M.D. 1985. Magnetic properties of iron in soil oxides and clay minerals. *In* J.W. Stucki, B.A. Goodman and U. Schwertmann (ed.) *Iron in soils and clay minerals*. Reidel. (In Press.)
- Curi, N. 1983. Lithosequence and toposequence of Oxisols from Goias and Minas Gerais States, Brazil. Ph.D. thesis. Purdue Univ., W. Lafayette, IN (Diss. Abst. Int. 44:1674-B).
- Curi, N., and D.P. Franzmeier. 1984. Toposequence of Oxisols from the Central Plateau of Brazil. *Soil Sci. Soc. Am. J.* 48:341–346.
- Davey, B.G., J.D. Russell, and M.J. Wilson. 1975. Iron oxide and clay minerals and their relation to colours of red and yellow Podzolic soils near Sydney, Australia. *Geoderma* 14:125–138.
- Dorr, J.V.N., and A.L.M. Barbosa. 1963. Geology and ore deposits of the Itabira District, Minas Gerais, Brazil. U.S. Geological Survey Professional Paper 341-C:C1–C108.
- EMBRAPA. Servico Nacional de Levantamento e Conservacao de Solos. 1985. Levantamento de reconhecimento de media intensidade dos solos e avaliacao da aptidao agricola das terras do Alto Paranaiba, MG. Empresa Brasileira de Pesquisa Agropecuaria. Ministerio da Agricultura, Rio de Janeiro. (In press.)
- Fontes, M.R., J.D. Fabris, M. Resende and E.G. Silva. 1985a. Concrecoes ferruginosas de alguns solos brasileiros. II. Caracterizacao mineralogica. *Revista Brasileira de Ciencia do Solo* 9:119–123.
- Fontes, M.R., M. Resende, and A.C. Ribeiro. 1985b. Concrecoes ferruginosas de alguns solos brasileiros. I. Caracterizacao quimica. *Revista Brasileira de Ciencia do Solo* 9:113–117.
- Gair, J.E. 1962. Geology and ore deposits of the Nova Lima and Rio Acima Quadrangles, Minas Gerais, Brazil. U.S. Geological Survey Professional Paper 341-A:A1–A65.
- Kabata-Pendias, A., and H. Pendias. 1984. Trace elements in soils and plants. CRC Press, Inc. Boca Raton, FL.
- Klug, H.P., and L.E. Alexander. 1974. X-ray diffraction procedures for polycrystalline and amorphous materials. 2nd ed. John Wiley & Sons, New York.
- Kullerud, G., R.M. Steffen, P.C. Simms and F.A. Rickey. 1979. Proton induced X-ray emission (PIXE)—A new tool in geochem-

- istry. Chem. Geol. 25:245-256.
- Lechler, P.J., W.R. Roy, and R.K. Leininger. 1980. Major and trace element analysis of 12 reference soils by inductively coupled plasma-atomic emission spectrometry. Soil Sci. 130:238-241.
- Lindsay, W.L. 1979. Chemical equilibria in soils. John Wiley & Sons, New York.
- Mehra, O.P., and M.L. Jackson. 1960. Iron oxide removal from soils and clays by a dithionite-citrate-bicarbonate system buffered with sodium bicarbonate. Clays Clay Minerals 7:317-327.
- Murphy, J., and J.P. Riley. 1962. A modified single solution method for the determination of phosphate in natural waters. Anal. Chim. Acta 27:31-36.
- Norrish, K., and R.M. Taylor. 1961. The isomorphous replacement of iron by aluminum in soil goethites. J. Soil Sci. 12:294-306.
- Rajan, S.S.S., and R.L. Fox. 1975. Phosphate adsorption by soils: II. Reactions in tropical acid soils. Soil Sci. Soc. Am. Proc. 39:846-851.
- Resende, M., D.P. Santana, D.P. Franzmeier, and J.M.D. Coey. 1986. Magnetic properties of Brazilian Oxisols. In Proc. of the Eighth Int. Soil Classification Workshop. U.S. Dep. Agric. Soil Management Support Services/Emperssa Brasileira de Pesquisa Agropecuaria (EMBRAPA)/Univ. of Puerto Rico, EMBRAPA, Rio de Janeiro, Brazil. (In press.)
- Resende, M. 1976. Mineralogy, chemistry, morphology and geomorphology of some soils of the Central Plateau of Brazil. Ph.D. Thesis, Purdue Univ., W. Lafayette, IN (Diss. Abstr. 37:4803).
- Santana, D.P. 1984. Soil formation in a toposequence of oxisols from Patos de Minas region, Minas Gerais State, Brazil. Ph.D. thesis, Purdue Univ., W. Lafayette, IN (Diss. Abstr. 45:1974-B).
- Schulze, D.G. 1981. Identification of soil iron minerals by differential X-ray diffraction. Soil Sci. Soc. Am. J. 45:437-440.
- Schulze, D.G. 1984. The influence of aluminum on iron oxides. VIII. Unit-cell dimensions of Al-substituted goethites and estimation of Al from them. Clays Clay Minerals 18:301-312.
- Schwertmann, U., E. Murad, and D.G. Schulze. 1982. Is there Holocene reddening (hematite formation) in soils of axeric temperate areas? Geoderma 27:209-223.
- Schwertmann, U., and R.M. Taylor. 1977. Iron oxides. p. 145-176. In Dixon, J.B. and S.B. Weed (ed.) Minerals in soil environments. Soil Science Society of America, Madison, WI.
- Sillanpää, M. 1982. Micronutrients and the nutrient status of soils: a global study. FAO Soils Bull. 48. Food and Agric. Organization of the United Nations, Rome.
- Simmons, G.C. 1968. Geology and iron deposits of the western Serra do Curral, Minas Gerais, Brazil. U.S. Geological Survey Professional Paper 341-G:G1-G53.
- Soil Survey Staff. 1975. Soil Taxonomy—A basic system of soil classification for making and interpreting soil surveys. Agriculture Handbook No. 436. SCS, U.S. Department of Agriculture.
- Steinwehr, H.E.v. 1967. Gitterkonstanten im System α -(Al,Fe,Cr)₂O₃ und ihr Abweichen von der Vegardregel. Z. Krist. 125:377-403.
- Syers, J.K., M.G. Browman, G.W. Smillie, and R.B. Corey. 1973. Phosphate sorption by soils evaluated by the Langmuir adsorption equation. Soil Sci. Am. Proc. 37:358-363.

ACKNOWLEDGMENT

The assistance of Dr. Mauro Resende in helping to plan the research, to collect the samples and to provide support in so many ways for the completion of this project is sincerely recognized.